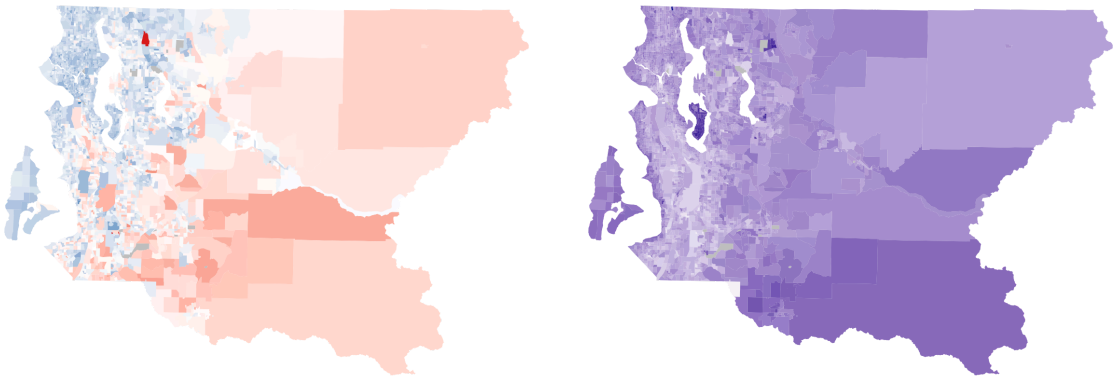
For this problem set, I used data from King County, Washington, which is home to the city of Seattle, where I am from. The first dataset I chose to dive into was focused on a mysterious King County Proposition 1, which I did not end up figuring out what it was. All I knew is the vote distribution of Approval and Disapproval of the proposition closely mirrored what I knew about Washington political ideologies, with Seattle and its direct surroundings voting a certain way, and the eastmost suburbs and districts voting the other. This led me to believe that King County Proposition 1 was favorable to Democratic voters. I plotted the data with first a choropleth to visualize broader geographical trends, and then again plotting voter share to see how voter turnout differed in different regions:

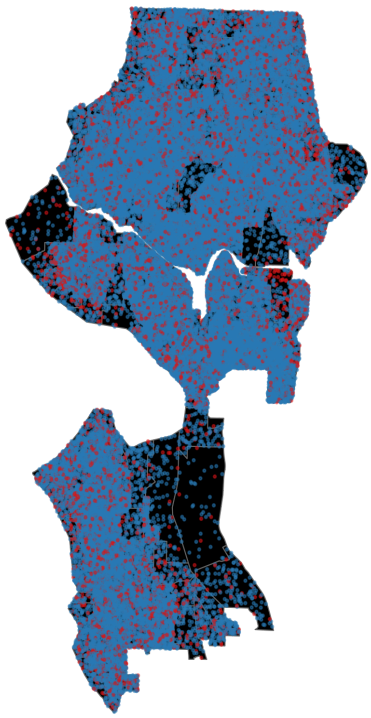


For the sake of simplicity in formatting, I excluded labels and legends this time. But I will explain that the left graph maps proposal approval share and the right voter share. As is evident, the eastmost regions of king county, far from any major cities or town centers, leaned towards disapproval. However, what strikes me as interesting is that those districts seemed to have the highest voter share and turnout. I realized that perhaps purely taking the proportion was not an accurate representation of the actual number of residents in a given district. Additionally, because I have no idea what King County Proposal 1 is, I decided to move to a different data and tackle the turnout in a slightly different manner.

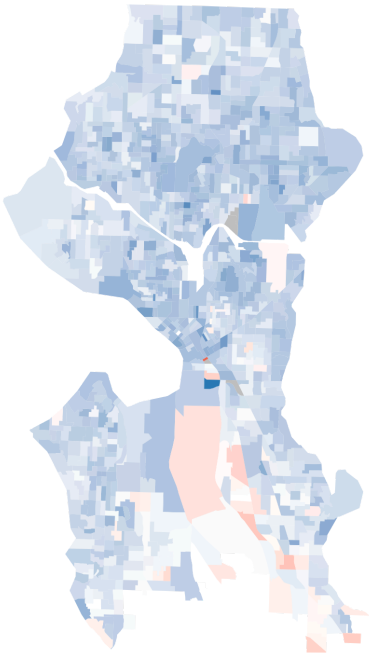
The dataset I chose to tackle next was the Congressional District 7 Representative Race, featuring incumbent Pramila Jayapal and opponent Dan Alexander. Jayapal prefers the Democratic Party, while Alexander the Republican.

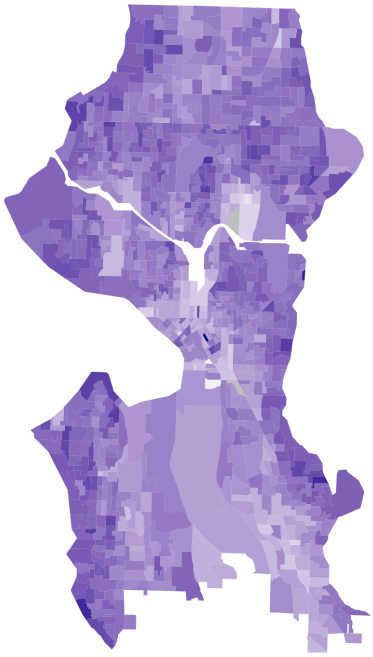
When plotting the initial choropleth of the Election results, it is relatively bland. As District 7 spans pretty much the city lines of Seattle, which is known to be a blue bubble, the results are unsurprising. Jayapal, who has served as a representative for Washington since 2017, won the race with 83% of the vote. As such, when mapping Jayapal's share of the vote, almost the entire map is dark blue, with only slight fluctuations scattered throughout. As such, I decided to go with the dot plot strategy, with each dot representing five specific voters. Using st_sample, this plot does a better job of including the voters who voted a different way. As we can see, there still exists a bit of red, albeit scattered pretty evenly throughout. I also thought this plot did a good job of highlighting the densely populated areas of Seattle. Notably, there are less residents in the south industrial district which is also where the large stadiums are.





As an additional exercise, I mapped the first dataset, King County Proposition 1, to just Seattle, to see how both plots would differ. However, I found them to be unassuming and uninteresting:





The only notable thing is that when zoomed in and compared with the Congressional Race dot plot, we can see that the largest red precinct for this proposal was indeed in that industrial district, where a significantly smaller population resides.